TD 2

Exercice 2:

1)

$Y=y$	-2	-1	0	1	2	3	Total
$P(Y=y)$	$\frac{32}{63}$	$\frac{16}{63}$	$\frac{8}{63}$	$\frac{4}{63}$	$\frac{2}{63}$	$\frac{1}{63}$	1

2) $S = |X+Y|$; $S \in \{0,1,2,3,4\}$

X \ S	0	1	2	3	4	Total
-2	$\frac{4}{378}$	$\frac{10}{378}$	$\frac{16}{378}$	$\frac{32}{378}$	$\frac{64}{378}$	$\frac{126}{378}$
1	$\frac{64}{378}$	$\frac{160}{378}$	$\frac{16}{378}$	$\frac{8}{378}$	$\frac{4}{378}$	$\frac{252}{378}$
	$\frac{68}{378}$	$\frac{170}{378}$	$\frac{32}{378}$	$\frac{40}{378}$	$\frac{68}{378}$	1

3)

S	0	1	2	3	4	Total
$P(S)$	$\frac{68}{378}$	$\frac{170}{378}$	$\frac{32}{378}$	$\frac{40}{378}$	$\frac{68}{378}$	1

4) Remarque: $P(X=x \cap Y=y) = P(X=x) \cdot P(Y=y)$ ⟶ X et Y indep

$P(X=-2 \cap S=0) = \frac{4}{378}$

$P(X=-2) = \frac{2}{6}$

$P(S=0) = \frac{68}{378}$

⟶ ~~$P(X=-2)$~~ $P(X=-2) \cdot P(S=0) = \frac{68}{378} \times \frac{2}{6} \neq \frac{4}{378}$

⇒ X et S sont ~~non~~ dépendantes.

Exercice DS 2022

1)

X	1	2	Tot
$P(X=x_i)$	$p_1 = 1/3$	$2p_1 = 2/3$	$1 = 3p_1$, $p_1 = \frac{1}{3}$

$Y=y$	0	1	Tot
$P(Y=y)$	$3/4$	$1/4$	1

<=Exercice 1 DS 2022

2) $P(X<1) = P(X \leq 0) = 0$

$P(Y \leq 0) = P(Y=0) = \frac{3}{4}$

3)

X / Y	1	2	Total
0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$
1	$\frac{1}{12}$	$\frac{2}{12} = \frac{1}{6}$	$\frac{1}{4}$
Tot	$\frac{1}{3}$	$\frac{2}{3}$	1

4) $P(X=2 / Y=1) = \dfrac{P(X=1 \cap Y=1)}{P(Y=1)}$

$= \dfrac{\frac{1}{12}}{\frac{1}{4}} = \frac{4}{12} = \frac{1}{3}$

5) $W = X + 2Y$.

$W \in \{1, 2, 3, 4\}$

W	1	2	3	4	Tot
$P(W=w)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{12}$	$\frac{1}{6}$	1

$P(W=1) = P(X=1) P(Y=0) = \frac{1}{4}$

$P(W=2) = P(X=2) P(Y=0) = \frac{1}{2}$

$P(W=3) = P(X=1) P(Y=1) = \frac{1}{12}$

$P(W=4) = P(X=2) P(Y=1) = \frac{2}{12} = \frac{1}{6}$

Exercice DS2021

<=Exercice 1 DS 2021

1- $a + 2a + 3a + 4a = 10a = 1$

$a = \frac{1}{10}$

2-

$X=x$	3	4
$P(X=x)$	$3a = \frac{3}{10}$	$7a = \frac{7}{10}$

$Y=y$	1	2
$P(Y=y)$	$4a = \frac{4}{10} = \frac{2}{5}$	$6a = \frac{6}{10} = \frac{3}{5}$

3) $P(X=3 \cap Y=1) = a = \frac{1}{10}$

$P(X=3) \times P(Y=1) = \frac{3}{10} \times \frac{2}{5} = \frac{6}{50} > \frac{1}{10}$

Donc X et Y sont non indépendantes

4) $P(X=x / Y=1) = P(X=3 / Y=1) \cup P(X=4 / Y=1)$

$= \frac{P(X=3 \cap Y=1)}{P(Y=1)} + \frac{P(X=4 \cap Y=1)}{P(Y=1)}$

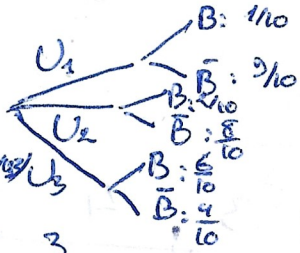
$= \frac{a}{4a} + \frac{3a}{4a} = \frac{1/10}{4/10} + \frac{3/10}{4/10}$

$= 1$

$X=x / Y=1$	3	4	Total
$P(X=x/Y=1)$	$1/4$	$3/4$	1

DS 2021 & 2022

Exercice 3 :

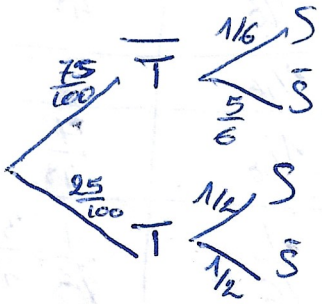


<=Exercice 3 DS 2021

$P(B) = P(U_1) \times P(B/U_1) + P(U_2) \times P(B/U_2) + P(U_3) \times P(B/U_3)$

$P(B) = \frac{1}{3} \times \frac{1}{10} + \frac{1}{3} \times \frac{2}{10} + \frac{1}{3} \times \frac{6}{10} = \frac{9}{30} = \frac{3}{10}$

Exercice 2 :



<= Exercice 2 DS 2022

~~P(S/T)~~

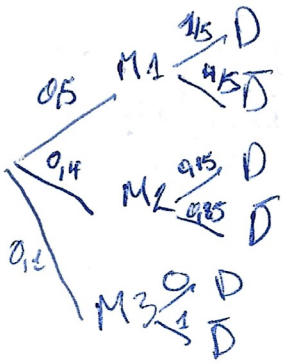
$P(T/S) = \frac{P(T \cap S)}{P(S)} = \frac{P(S/T) \times P(T)}{P(S)}$

$P(T/S) = \frac{1/2 \times \frac{25}{100}}{1/2 \times \frac{25}{100} + 1/6 \times \frac{75}{100}} = \frac{1}{2}$

$P(M1/D) = \frac{P(M1 \cap D)}{P(D)} = \frac{P(D/M1) \times P(M1)}{P(D)}$

<= Exercice 3 DS 2022

$= \frac{1/5 \times \frac{1}{2}}{1/5 \times \frac{1}{2} + 0,15 \times 0,4}$



$= \frac{\frac{1}{10}}{\frac{1}{10} + 0,06} = 0,625$

TD2

Exercice 3

1)

$X=x_i$ / $Y=y_j$	1	2	3	4	Total
1	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{12}$	$\frac{1}{16}$	$\frac{25}{48}$
2	–	$\frac{1}{8}$	$\frac{1}{12}$	$\frac{1}{16}$	$\frac{13}{48}$
3	–	–	$\frac{1}{12}$	$\frac{1}{16}$	$\frac{7}{48}$
4	–	–	–	$\frac{1}{16}$	$\frac{1}{16} = \frac{3}{48}$
Total	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	1

2) $P(X=Y) = P(X=1, Y=1) + P(X=2, Y=2) + P(X=3, Y=3) + P(X=4, Y=4)$

$= \frac{1}{4} + \frac{1}{8} + \frac{1}{12} + \frac{1}{16} = \frac{25}{48}$

~~$P(X=1) P(Y=1) + P(X=2) P(X=2) + P(X=3) P(X=3) + P(X=4) P(X$~~

3)

$Y=y_j$	1	2	3	4	Total
$P(Y)$	$\frac{25}{48}$	$\frac{13}{48}$	$\frac{7}{48}$	$\frac{3}{48}$	1

Exercice révision

1.) $E_1 \cap E_2 = A \cap \bar{B} \cap \bar{C} \cap A \cap (B \cup C) = A \cap \bar{B} \cap \bar{C} \cap (B \cup C)$

$= A \cap \bar{B} \cap \bar{C} \cap B \cup A \cap \bar{B} \cap \bar{C} \cap C$

$\emptyset \cup \emptyset = \emptyset$

ou bien

$\overline{B \cup C} = \bar{B} \cap \bar{C}$

$A \cap (\overline{B \cup C}) \cap (B \cup C) = \emptyset$

⇒ Donc E_1 et E_2 sont incompatibles.

2.) $E_1 \cup E_2 = E_1 + E_2 - P E_1 \cap E_2 = E_1 + E_2$

$= A \cap \bar{B} \cap \bar{C} \cup A \cap (B \cup C)$

$= A \cap (\bar{B} \cap \bar{C} \cup \overline{\bar{B} \cap \bar{C}})$

$= A \cap \Omega = A$

3.) $P(E_1) = P(A \cap \bar{B} \cap \bar{C}) = 0,35$ ($E_1 \cup E_2 = A \Rightarrow P(E_1 \cup E_2) = P(A)$, $P(E_1) \cup P(E_2) - P(E_1 \cap E_2) = 0,6$)

$P(E_2) = P(A \cap (B \cup C)) = P((A \cap B) \cup (A \cap C)) = P(A \cap B) + P(A \cap C) - P(A \cap B \cap A \cap C)$

$= 0,2 + 0,1 - 0,05 = 0,25$ ($A \cap B \cap C$)

$=P(A) - P(E_1)$

4) A et B sont indépendants?

E1 et E2 ?

$P(E1 \cap E2) = 0 \neq P(E1) \times P(E2) = 0,25 \times 0,35$

$P(A \cap B) = 0,2 \neq P(A) \times P(B) = 0,6 \times 0,4$

⇒ A et B ne sont pas indépendantes

⇒ E1 et E2 ne sont pas indépendantes

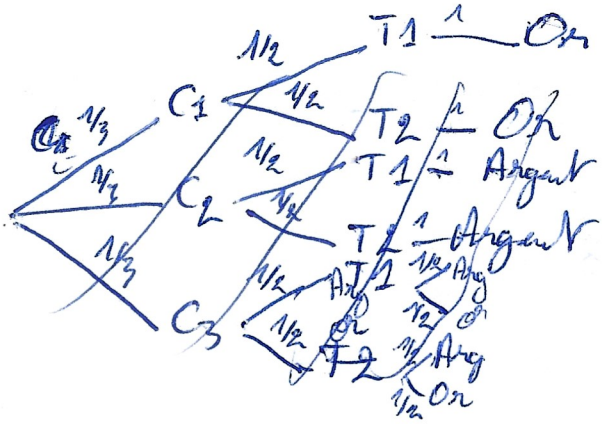
A et C sont indépendantes?

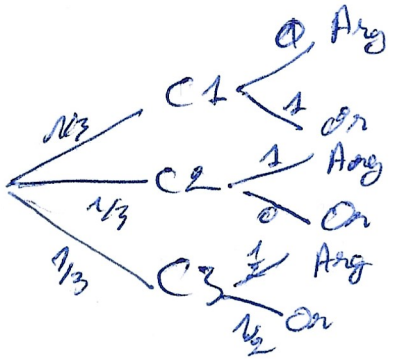
B et C sont indépendantes?

$P(A \cap C) = 0,1 \neq P(A) \times P(C) = 0,6 \times 0,3 = 0,27$

$P(B \cap C) = 0,1 \neq P(B) \times P(C) = 0,4 \times 0,3 = 0,12$

Exercice 2 révision :





1) $P(C2/A) = \frac{P(C2 \cap A)}{P(A)} = \frac{P(A/C2) \times P(C2)}{P(A)} = \frac{1 \times \frac{1}{3}}{\frac{1}{3} \times (\frac{1}{2}+\frac{1}{2}) + \frac{1}{3} \times \frac{1}{2}(\frac{1}{2}+\frac{1}{2})}$

$= \frac{P(A/C2) \times P(C2)}{P(C1)P(A/C2) + P(A/C1) \times P(C1) + P(C3) \times P(A/C3)} = \frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{3} \times \frac{1}{2}} = \frac{2}{3}$

$= \frac{1/3}{\frac{1}{3} \cdot 0 + \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot \frac{1}{2}} = \frac{1/3}{1/2} = \frac{2}{3}$

2) $P(N) = C2\,C2 + C3\,C3 + \underbrace{C1\,C1}_{0}$

$P(N) = \frac{2}{3} \times \frac{2}{3} + \left(\frac{1/6}{1/2}\right) \frac{1}{6}$

$= \frac{2}{9} + \frac{1}{9} = \frac{3}{9} = \frac{1}{3}$ (Indépendamment)